

Uzair Khan

Network Operations Engineer · Satellite NOC Specialist

Manassas, VA

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CompTIA CySA+ (CS0-003, through 2027)

SUMMARY

Network Operations Engineer with 8+ years in satellite and enterprise NOC environments. **Hands-on iDirect platform experience** across GEO and MEO (O3b/SKALA) constellations — terminal restoration, beam switching, P1dB test coordination, commissioning, and real-time capacity management. Proven in 24/7 Tier 2/3 NOC operations supporting maritime vessels, aero/inflight connectivity, and fixed VSAT for global satellite operators. Experienced in BGP/OSPF operations, Juniper MX5/EX, Cisco ASA, and Splunk SIEM. Strong incident command, RCA documentation, and ServiceNow lifecycle management.

iDirect

O3b MEO / SKALA

DVB-S2 / ModCod

P1dB Testing

Beam Switching

VSAT Commissioning

BGP / OSPF

Juniper MX5

Cisco ASA

Splunk SIEM

SolarWinds

ServiceNow

NOC Specialist (Satellite)

2026 – Present

Apex Systems, contractor to AT&T Global Public Sector · Mount Weather Emergency Operations Center, Bluemont, VA

- **Tier 1 NOC:** Monitor, troubleshoot and configure the DHS/FEMA voice, data and satellite network on a weekend and after-hours rotation.
 - **Satellite Hub Operations:** Operate earth station equipment at an iDirect satellite hub supporting FEMA mobile units; change space-segment bandwidth assignments for new and changed mission requirements.
 - **Reporting:** Produce bandwidth usage, traffic analysis and daily schedule reports for the government task order.
 - **Remote Access:** Maintain remote access servers and remote client systems, including software patches and updates.
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IT Functional Analyst II / Systems Administrator

2024 – 2025

MDThink – DoIT: Maryland Benefits (SNAP/DHS) · Linthicum, MD

- **Endpoint Fleet Management:** Managed 200+ Windows 10/11 devices using Microsoft Intune, SCCM, and Autopilot. Administered Azure AD/Entra ID conditional access, group-based licensing, and MFA enforcement.
 - **Asset Inventory Automation:** Built Python/PowerShell scan script to inventory state-owned and vendor devices; automated reporting to Confluence and distributed to VP and management.
 - **SOP Authoring:** Authored enterprise KB articles and SOPs covering endpoint lifecycle workflows adopted organization-wide.
 - **AI Chatbot:** Prototyped AI-powered IT helpdesk chatbot for Google Spaces to reduce tier-1 ticket volume.
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SES Networks · Manassas, VA

Tier 2/3 NOC operations for a global satellite and IP network serving maritime vessels, fixed VSAT, and aero (inflight connectivity) across GEO and MEO constellations. Primary NOC contact for SKALA (O3b MEO) customers and Virtual Network Operators (VNOs).

- **Terminal Restoration & Configuration:** Logged directly into iDirect, Hughes, and Comtech terminals via in-band and out-of-band redundant connections (management VLAN, cellular backup, Cisco ASA VPN) to clear equipment faults, push configuration changes, and restore customer traffic. Executed port forwarding corrections and parameter adjustments during active service-affecting events.
- **iDirect Platform Operations:** Administered terminals daily via iVantage NMS and iQ Desktop — monitored Eb/No, Ec/No, RX/TX power levels, ACM state, symbol rates, BER, and burst timing. Pulled and analyzed modem logs (network_entry, burst_timing, demod_lock) to isolate RF vs. network layer faults.
- **Beam Switching & Capacity Management:** Executed manual beam switches for maritime vessels via GUI pre-config profiles and direct CLI on iDirect modems. Monitored beam utilization across the fleet and rebalanced vessel distribution to resolve congestion — coordinating with capacity planning on significant migrations.
- **SKALA / O3b MEO Operations:** Managed bandwidth package changes for SKALA MEO customers via GUI and CLI; monitored MEO satellite handoffs; intervened on failed handoff events. Acted as primary NOC contact for SKALA VNO service events on the O3b MEO constellation.
- **DVB-S2 / RF Management:** Performed real-time ModCod adjustments on DVB-S2 forward and return carriers. Differentiated rain fade from RFI and coordinated appropriate response — including spectrum analysis escalations to RF engineering.
- **P1dB Test Coordination:** Coordinated and bridged P1dB transmit power tests for SKALA, VSAT, and VNO customers. Managed call bridge between site technician, ground station RF engineers, and NOC — stepped through power increments, documented compression points, locked approved TX ceilings into NMS terminal profiles, and cleared carriers for operational use.
- **Site Commissioning & Recommissioning:** Commissioned new fixed and maritime VSAT terminals end-to-end: alignment verification, modem registration, link margin validation, P1dB test, service activation, as-built documentation. Recommissioned sites after equipment swaps and vessel dry-dock returns.
- **Root Cause Investigation:** Conducted historic RCA investigations by reviewing configuration changes, beam architecture modifications, local interference sources, and modem logs. Coordinated remote technician site visits for physical layer fault resolution. Authored structured RCA reports in ServiceNow.
- **Network Layer Operations:** Monitored OSPF backbone links and escalated path failures with structured summaries. Performed BGP diagnostics on Juniper MX5 vessel CPE — neighbor state, prefix advertisements, session reset analysis. Used Cisco ASA for out-of-band management access to dark remote sites.
- **Splunk SIEM:** Onboarded 30+ log sources; built SPL searches and threshold alerts for recurring fault patterns, reducing MTTD across the environment.
- **Incident Command:** Led incident bridges during service-affecting events, coordinating across satellite ops, terrestrial engineering, ground station teams, and vendor NOCs. Authored post-incident RCA reports and maintained shift handover documentation.
- **Training & Documentation:** Created commissioning runbooks, P1dB test procedures, and troubleshooting flowcharts. Mentored junior NOC analysts on iDirect log analysis and escalation procedures.

Impact: Maintained 99.9%+ availability across maritime and aero links. Splunk integration improved fault correlation across 30+ nodes. P1dB bridge coordination reduced SKALA and VNO customer activation delays.

Enterprise Network Technician — Network Operations Center

2021 – 2022

ViaPath · Falls Church, VA

- **WAN/LAN Monitoring:** Monitored infrastructure across 100+ correctional facility sites using SolarWinds NPM/NTA; responded to SNMP-triggered alerts within 15-minute SLA windows.
- **Incident Triage:** Triageed events using IR playbooks; coordinated with on-site contacts and LECs to validate circuit-level disruptions and maintenance windows.
- **Satellite Failover:** Initiated ViaSat satellite failover circuits for remote sites; documented activation and restoration procedures per site runbook.
- **Change Management:** Tracked planned maintenance through structured CR processes; cross-referenced schedules with live alerts to suppress false-positive escalations.

Impact: Improved uptime across mission-critical correctional infrastructure through rapid SLA response and structured carrier coordination.

Enterprise Account & Operations Support

2017 – 2020

Carfax · Centreville, VA

- Tier 2 support for CARFAX RESTful API and POS software integrations. Diagnosed OAuth/authentication issues between dealer management systems and vehicle history APIs.
- Managed ticket lifecycle in Salesforce CRM; collaborated with engineering on structured API defect escalations with reproduction steps and log artifacts.

Retail Operations Manager

2012 – 2016

AT&T Mobility · Oakton, VA

- Operated AT&T OPUS NCR tablet POS and backend CLI provisioning system for wireless service activation, SIM configuration, and provisioning failure resolution at carrier backend level.
- MDM enrollment for business accounts; enterprise email, VPN, and MAM policy configuration. Led team of 10.

TECHNICAL SKILLS

Satellite / RF	iDirect (iVantage, iQ Desktop), O3b MEO/SKALA, Hughes, Comtech CDM series, DVB-S2, SCPC/MCPC, ModCod/ACM, P1dB testing, beam switching, VSAT commissioning, HTS/FSS, aero IFC, maritime NOC, satellite handoffs
Networking	OSPF, BGP, EIGRP, MPLS, VLANs, STP, LACP, DNS, DHCP, NTP, VPN, LAN/WAN/MAN, Juniper MX5/EX (JunOS), Cisco ASA/IOS/NXOS
Monitoring	SolarWinds NPM/NTA, Splunk SIEM, SNMP, NetFlow, syslog, Wireshark, tcpdump, NMS (multi-vendor)
Automation	Python, PowerShell, bash scripting, Splunk SPL
ITSM	ServiceNow, Jira (JQL, dashboards), Confluence, Google Workspace
Documentation	As-builts, SOP/KB authoring, RCA reports, CR management, commissioning runbooks
Endpoint / MDM	Microsoft Intune, SCCM, Autopilot, Azure AD/Entra ID, Active Directory, GPO

PROJECTS

Home Network Monitoring Lab (Raspberry Pi)

Enterprise-grade home NOC: 15 Docker services — Grafana, Prometheus, Cowrie SSH honeypot, Pi-hole + Unbound DNS (DNSSEC), fail2ban, Tailscale VPN, ntfy alerting. Security dashboards, nmap device discovery; led a capacity-driven incident and migration plan.

TankPit Discord Bot & RAG AI System · github.com/uk0430

Full-stack Discord bot with hybrid RAG + SQL AI engine. Local LLM inference (Mistral 7B / llama.cpp), nomic-embed vector store, 13,250-player SQLite database. Python, asyncio, Docker.

EDUCATION & CERTIFICATIONS

CompTIA CySA+ (CS0-003)

Valid through 2027

A.A.S. Information Systems Technology (in progress)

Northern Virginia Community College · 2026

Security Clearance

Public Trust (Top Secret in process)

A.A.S. Business Administration

Northern Virginia Community College